CLIENT REPORT

Home Depot vs Lowe's: Search Relevance Discoveries

An LLM-judged relevance benchmark of Home Depot and Lowe's on-site search across three query datasets and 900+ graded product judgments — showing where each engine wins, breaks, and loses shoppers.

Prepared for

Home Improvement Retail Search Benchmark

Prepared by Northvalley Intelligence LLC

REPORT PROMISE

Decision-ready findings, practical priorities, and a clear path from review to implementation.

FORMAT

Multi-page

Cover, executive summary, findings, roadmap, and next steps.

Executive Summary

THE HEADLINE

Home Depot and Lowe's are near-tied on core relevance, but they fail in opposite, fixable ways. Lowe's wins on synonyms; Home Depot edges project search; and both mishandle industry specs — Home Depot ignores them, Lowe's over-reacts to them.

We evaluated live on-site search on **homedepot.com** and **lowes.com** across three query datasets: (1) **synonym/category** equivalence, (2) **dimensions** — measurement notation and the "#" spec gradations, and (3) **pro/job-function** project and trade queries. Results were captured by bounded, same-origin browser reads (no crawling or scraping); result counts read deterministically; and each returned product graded 0–3 for relevance by a local LLM judge, scored as NDCG. The three headline numbers are at right.

DIMENSIONS NDCG
0.90 / 0.88
Home Depot vs Lowe's — near tie on core relevance.

SYNONYM MATCH
30 / 48%
Lowe's returns consistent results for more equivalent terms.

PROJECT NDCG
0.83 / 0.78
Home Depot edges project and trade queries.

What This Answers

- Which engine returns more relevant results, and where does each break?
- Do the engines understand industry specs ("#57 stone", "#4 rebar") and notation ("1/2-in")?
- Which is better for natural-language project shopping ("build a deck")?

Key Findings

FINDING 01

Industry specs are the biggest gap

The "#" gradations (#57 stone, #4 rebar, #7 drill bit) are hard catalog specs. **Home Depot ignores them** — it returns generic in-category products and never surfaces the true spec. **Lowe's over-reacts** — it treats the "#" as a different query and often derails to the wrong category entirely. Neither reliably finds the exact spec'd item.

FINDING 02

Notation fails oppositely

Home Depot breaks on **hyphenation**: "1/2-in screw" drops about 70% of the results of "1/2 in screw" — a recall cliff. Lowe's breaks on **unit words**: "10 foot cable" returns a fraction of "10 ft cable". Relevance of what each keeps stays high, so these are recall failures, not ranking failures.

FINDING 03

Project search favors Home Depot

Both return real kits for most projects. Home Depot edges it and **monetizes "install a..."** queries as paid services; a few Lowe's project queries (kitchen faucet, toilet, paint a room) return little. Both fail bare trade names — "electrician" and "landscaper" return **zero products**.

BUSINESS IMPLICATION

These are not vague "search feels off" complaints — each is a specific, reproducible query behavior with a clear fix. Spec parsing and notation normalization are self-contained engine changes; synonym coverage is a mapping problem. All three convert directly into found products, fewer dead-ends, and larger baskets.

Evidence Snapshot

Query (retailer)	What we observed
#57 stone (Lowe's)	Window blinds and a quartz sink; "57 stone" returns real #57 aggregate.
#7 drill bit (Lowe's)	Magnetic drill presses; "7 mm drill bit" returns real twist bits.
#4 rebar vs 4 rebar (HD)	Near-identical sets — the "#" spec is ignored.
1/2-in screw (HD)	Drops about 70% vs "1/2 in screw" (hyphen recall cliff).
replace a kitchen faucet (Lowe's)	5 results, a stray wire listing; HD returns real faucets.

Recommended Improvements

The findings translate into a clear priority sequence. Each fix targets a specific, reproducible query behavior — ordered by shopper impact and engineering self-containment.

Phase	Focus	Recommended work	Outcome
Now	**Spec parsing (both)**	Treat "#57", "#4", "#7" as catalog gradations, not noise; for Lowe's, stop the "#" from derailing the query.	Finds the exact SKU
Next	**Notation normalize**	Make hyphen / unit-word / decimal / quote variants equivalent ("1/2-in" = "1/2 in" = "0.5 in").	No recall cliffs
Then	**Synonyms (esp. HD)**	Map equivalents and brands: drywall/sheetrock, weed killer/roundup, concrete/quikrete.	Fewer dead-ends
Later	**Project kits**	Return complete project kits for natural-language jobs; disambiguate bare trade names.	Larger baskets

PER-RETAILER PRIORITY

Home Depot: spec awareness (the "#" is ignored) and synonym coverage (30% vs 48%). **Lowe's:** stop the "#" from derailing to the wrong category, and shore up a few empty project queries.

HOW THIS WAS MEASURED

Live search on both sites via bounded same-origin browser reads; deterministic result counts; every returned product graded 0–3 by a local LLM judge (auditable per-item traces) and scored as NDCG; plus count-ratio and SKU-overlap checks. Repeatable and self-verifiable.

Roadmap and Next Steps

WHERE THIS GOES NEXT

The evaluation harness is live and repeatable — three datasets, both retailers, LLM-judged NDCG, drillable to per-product grades. The next axis is recommendation quality: are a product's suggested add-ons the ones a shopper would actually buy alongside it?

TRUST AND AUDITABILITY

Every grade behind every score is a stored trace with the exact prompt and model output — a questionable number can be inspected, not just trusted. The judge was validated against a hand-labelled gold set, and two notation defects were caught and fixed before scoring.

Extend the Benchmark

- Complementary-recommendation quality (does a drill suggest bits, batteries, a case?)
- Full transformation-type coverage across every dimension seed
- Additional retailers and departments under the same rubric
- Track scores over time as each engine ships fixes

At a Glance

- Synonyms: Lowe's ahead (48% vs 30%)
- Dimensions relevance: tie (0.90 / 0.88), opposite spec failures
- Project / trade: Home Depot edge (0.83 / 0.78)
- 900+ graded judgments, fully auditable

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